MIL-W-27895A(USAF)
14 May 1965
SUPERSEDING
MIL-W-27895(USAF)
31 August 1962

MILITARY SPECIFICATION

WINCH, AIRCRAFT MOUNTED HCU-9/A

1. SCOPE

1.1 This specification covers 400-cycle, 115/200-volt, 3-phase, single speed, 6,500-pound capacity, portable aircraft cargo winch, designated HCU-9/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-P-116	Preservation, Methods of
MIL-W-1511	Wire Rope, Steel (Carbon) Flexible, Preformed
MIL-E-4970	Environmental Testing, Ground Support Equipment, General Specification for
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance
MIL-M-7969	Motors, Alternating Current, 400-cycle 115/200-Volt System, Aircraft, Class A and B, General Specification for
MIL-S-10379	Suppression, Radio Interference, General Requirements for Vehicles (and Vehicular Subassemblies)
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load-Bearing Base
MIL-R-27542	Reliability Program for Systems, Subsystems, and Equipment

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS33543	Criteria-Temperature and Altitude Range Self Cooled Electric Equipment

DRAWINGS

United Shoe Machinery Co.

73-20-100	Winch, Aircraft Mounted, HCU-9/A

FSC 1680

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73-20-119	Power Cable Assembly, C133 Aircraft
73-20-138	Power Cable Assembly, C130E Aircraft
73-20-139	Power Cable Assembly, C141 Aircraft
73-49-027	Adapter, Power Cable
73-71-003	Motor Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The winch shall consist of the following components:

Description	Requirement
Motor	3.5.1
Cable Drum Assembly	3.5.2
Limiting and Safety Devices	3.5.3
Wire Rope Assembly	3.5.4
Snatch Block	3.5.5
Control Cable and Control Product	3.5.6
Power Cables	3.5.7

3.3 Design and construction. Winches, components, assemblies and subassemblies shall be fabricated and assembled into a complete winch in accordance with Drawing 73-20-100.

3.3.1 Reliability. The winch shall demonstrate a reliability of 99.5 percent for an average mission time of 15 minutes with a minimum acceptable mean-time-between failures (MTBF) of 50 hours. Reliability shall be demonstrated to a 90 percent confidence level.

3.4 Performance. The winch shall be capable of satisfactorily withstanding and operating under the following conditions:

3.4.1 Rated capacity. The winch shall be capable of exerting 6,500 pounds on a single purchase line at a constant speed of 30 fpm plus or minus 15 percent in both the reel-in and unreel phase.

3.4.2 Environmental. The environmental conditions shall be as follows:

a. Temperatures ranging from minus 65° to plus 160°F

b. Relative humidity up to 100 percent

c. Exposure to salt-fog atmosphere

d. Sand and dust as encountered in desert areas

e. Explosive atmosphere as encountered in vapor within and surrounding the equipment.

3.5 Details of components

3.5.1 Motor. The motor (Drawing 73 71 003) shall be a single-speed, continuous-duty type conforming to MIL-M-7969, class A. The motor shall be explosion proof, fully reversible and shall operate on a 200 volt-to-line, or 115 volt-to-neutral, nominal voltage limits 105 to 122, 400-cycles, 3-phase, 4-wire. Voltage shall be in

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accordance with MIL-STD-704. The motor shall have sufficient starting torque to operate under full load with across-the-line starting.

3.5.1.1 Altitude temperature range. The altitude temperature range shall be in accordance with curve II of MS33543 from 0 to 5000 feet.

3.5.1.2 Thermal protection. The motor shall be provided with thermal protection as specified in MIL-M 7969, method II. The means of protection used shall be of the type that will provide optimum operational life consistent with maximum power capability of the motor under all operable conditions.

3.5.1.3 Running and starting current. The maximum running current at full load shall not exceed 30 amps per phase. The maximum starting current at full load shall not exceed 125 amps per phase.

3.5.2 Cable drum assembly. The cable drum assembly including level wind mechanism shall perform as follows under all operating and environmental conditions.

3.5.2.1 The manually-operated drum-disengaging mechanism shall operate by a manual control on the left side as viewed by the cable exit end which shall allow free manual unreeling of cable at cable speeds of up to 200 fpm with a line pull not to exceed 20 pounds.

3.5.2.2 The level wind mechanism shall prevent spooling and excessive slack in the wound cable when the cable is being power reeled in or out under no load, or being pulled from the disengaged drum. The level wind mechanism shall also operate in the above manner, under all loading conditions and with a fleet angle of 5 degrees on either side of a vertical, longitudinal plane which intersects the level wind fair lead.

3.5.3 Limiting and safety devices. The limiting and safety devices shall operate as follows:

3.5.3.1 Overload protection. The overload protection device shall operate at three cable tension settings which are to be adjusted manually. This device will automatically limit the line pull in the first setting between 4,000 and 4,500 pounds; in the second setting between 5,600 and 6,600 pounds; and the third setting between 7,000 and 8,500 pounds. The winch shall maintain the line pull that activated the overload device throughout the overload period.

3.5.3.2 Overwind and unreeling. Limiting devices shall operate in such a way as to prevent the winch from reeling in and unreeling beyond the provisioned limits.

3.5.3.3 Power failure. In event of power interruption to the winch, including pendant controlled stopping, the winch shall stop and maintain the line pull in effect at time of power interruption.

3.5.4 Wire rope assembly. The wire rope shall conform to MIL-W-1511. There shall be 200 feet of usable wire rope with hook incorporated in the winch assembly.

3.5.5 Snatch block. The snatch block assembly shall have a working capacity of 13,000 pounds.

3.5.6 Control cable and control pendant. The control cable and control pendant shall conform to applicable drawings except that a hook shall extend from the handle end of the pendant. This hook shall be

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metal,covered with the same material as the pendant body. The hook shall be used to hang the pendant on appropriate protrusions in the aircraft and will be of approximately 1 1/4 inch inside diameter.

3.5.7 Power cables. When specified, a combination of the following power cables shall be provided with each winch (see 6.2).

3.5.7.1 C-133A and B aircraft. The winch power cable for use on the C-133 A and B aircraft shall be in accordance with Drawing 73 20 119.

3.5.7.2 C-130E aircraft. The winch power cable for use on the C-130E aircraft shall be in accordance with Drawing 73 20 138.

3.5.7.3 C-141 aircraft. The winch power cable for use on the C-141 aircraft shall be in accordance with Drawing 73 20 139.

3.5.7.4 Adapter cable ground power. A power cable adapter shall be utilized for use of the winch connected directly to a ground power unit and shall be in accordance with Drawing 73 49 027.

3.6 Radio interference suppression. The winch shall meet radio interference requirements of MIL-S-10379. Measurements shall be made at a distance of 30 feet with power cable Drawing 73 20 139 in use.

3.7 Lubrication. Lubricants used in the winch shall conform to MIL-H-5606 and MIL-G-23827 and shall be suitable for all operating conditions specified herein.

3.8 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.9 Workmanship. The winch, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts and tightness of assembly screws and bolts, et cetera.

3.9.1 Cleaning. The winch shall be thoroughly cleaned of loose, spattered or excess solder, metal chips and other foreign material during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary by the Government to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the winch shall be classified as follows:

a. Preproduction testing
b. Acceptance tests.

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4.3 Test condition

4.3.1 Apparatus. Insofar as practicable, the apparatus used in conjunction with the testing specified herein shall be of the laboratory precision type and shall be calibrated at intervals properly spaced to insure laboratory accuracy.

4.4 Preproduction testing

4.4.1 Test sample. The test samples shall consist of two winches designated (a) and (b) representative of the production equipment. The winches shall be fully representative of winches to be supplied by the manufacturer from production facilities and tooling.

4.4.2 Preproduction tests. The preproduction tests shall consist of the following tests described under 4.6.

Winch (a)	Winch (b)
4.6.1	4.6.1
4.6.2	4.6.2
4.6.4	4.6.3
4.6.4.1	4.6.5
4.6.4.2	4.6.6
4.6.4.3	4.6.7
4.6.4.4	4.6.8
4.6.4.5	4.6.9
4.6.4.6	
4.6.9	

4.5 Acceptance tests. Acceptance tests shall consist of individual tests.

4.5.1 Individual tests. Each winch shall be subjected to the following tests as described under 4.6.

a. Examination of product
b. Functional tests

4.6 Test methods

4.6.1 Examination of product. The winch shall be inspected to determine compliance with the requirements specified herein with respect to completeness of components, materials, weight, workmanship, dimensions, clearances and marking.

4.6.2 Functional tests. The functional operation of each winch shall be tested as follows:

4.6.2.1 With the winch secured by two tiedown fittings, the winch shall be loaded in such a way as to assure that the overload safety device settings operate correctly. Activation of the overload shall occur with the first setting between 4,000 and 4,500 pounds; with a second setting between 5,600 and 6,600 pounds; and a third setting between 7,000 and 8,500 pounds. Upon activation, and at that line pull, the load shall maintain its position. The load position hold with power off shall also be demonstrated.

4.6.2.2 With a 6,500-pound line pull, 5 feet of cable shall be reeled in, then unreeled five times at least twice with a 5-degree fleet angle to the right and at least twice with a 5-degree fleet angle to the left with line pull at one end of the drum. Another 5-foot operation shall be performed with the line pull at the opposite end of the drum.

4.6.2.3 The manually-operated drum disengaging mechanism shall be activated and the full length of cable manually unreeled. Under these conditions, the cable shall not spool and unreeling force shall not exceed 25 pounds at any cable speed up to 200 fpm.

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4.6.2.4 The cable limit device at each end of the cable wind shall be checked to determine that the overwind limit devices function properly.

4.6.2.5 Snatch block test. The snatch block assembly shall be connected in double purchase and loaded to 13,000 pounds. While maintaining these load conditions the cable shall be retracted through the block. The snatch block shall operate smoothly and there shall be no failure or permanent deformation.

4.6.3 Level wind test. With a 6,500-pound line pull and the cable in the maximum fleet angle positions to the right and left, 100 feet of cable shall be reeled in and unreeled 20 times; ten pulls with maximum angle to the right and 10 pulls with maximum angle to the left. Two additional pulls shall be accomplished at no load and no angle. Under all these conditions the cable shall wind level with no spooling or excessive wear to the winch or cable.

4.6.4 Environmental tests. The winch shall be subjected to the following environmental tests in accordance with MIL-E-4970. Paragraph entitled "Performance Record" of MIL-E-4970 shall not apply.

4.6.4.1 Low temperature test. The entire winch including controls shall be subjected to a low temperature test in accordance with procedure I of MIL-E-4970, except that the tests shall be conducted at -65°F. During the tests, while at the specified temperature and with a 6,500-pound line pull applied, the winch shall be operated through 10 cycles consisting of unreeling 5 feet of cable, stopping, then reeling in the cable. Under these conditions, the device that stops the winch in the fully reeled position shall be checked for proper operation. At the completion of this cycling, all overload safety devices shall be checked for operation between specified limits (see 3.5.3.1). Power shall be disconnected and compliance with 3.5.3.3 checked. The full length of cable shall then be unreeled and the limiting device for cable unreeling checked.

4.6.4.2 High temperature test. The winch shall be subjected to the high temperature test in accordance with procedure I of MIL-E-4970. During this test, while at the specified temperature and with a 6,500-pound line pull applied to the cable, the winch shall be operated through 25 cycles consisting of unreeling 5 feet of cable, stopping, then reeling in the cable. Under these conditions, the device that stops in the fully reeled position shall be checked for proper operation. At the completion of this cycling, the overload safety device shall be checked for compliance with 3.5.3.1. Power shall be disconnected and compliance with 3.5.3.3 shall be checked. The full length of cable shall then be unreeled and compliance with 3.5.3.2 shall be checked.

4.6.4.3 Humidity test. The winch shall be subjected to a humidity test in accordance with procedure I of MIL-E-4970. At the completion of the exposure and at the specified environment, the winch shall be subjected to the tests specified in 4.6.2, except that the 10 cycles in 4.6.2.2 may be taken at any place on the drum and 4.6.2.5 shall not be accomplished.

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4.6.4.4 Salt-fog test. The winch shall be subjected to salt-fog specified in procedure I of MIL-E-4970. At the completion of the exposure, the winch shall be subjected to the tests specified in 4.6.2, except that the 10 cycles in 4.6.2.2 may be taken at any place on the drum and 4.6.2.5 shall not be accomplished.

4.6.4.5 Sand and dust test. At the completion of the salt-fog test, the winch shall be subjected to the sand and dust test in accordance with procedure I of MIL-E-4970. At the completion of the exposure, the winch shall be subjected to the tests specified in 4.6.2, except that the 10 cycles in 4.6.2.2 may be taken at any place on the drum and 4.6.2.5 shall not be accomplished.

4.6.4.6 Explosionproof tests. The winch shall be subjected to the explosionproof test in accordance with procedure III of MIL-E-4970.

4.6.5 Radio interference suppression test. The winch shall be operated and compliance with MIL-S-10379 shall be determined. Measurements shall be made at a distance of 30 feet. Power cable in accordance with Drawing 73 20 139 shall be used during this test.

4.6.6 Partial life test. With a minimum ambient temperature of 65°F., the winch shall be subjected to and shall operate without operational irregularity throughout a 200-cycle partial life test. Each cycle shall consist of the following:

a. 6,500-pound line pull. Reel in 100 feet of cable

b. 6,500-pound line pull. Unreel 100 feet of cable.

The first 100 cycles shall be run with the first 100 feet of cable on the drum. The second 100 cycles shall be run with the second 100 feet of cable on the drum. During this test, running and starting current shall be measured. Running current shall not exceed 30 amp per phase. Starting current shall not exceed 125 amp per phase.

4.6.7 Inching test. With a power source of 400-cycle, 200-volt ac, and the surrounding air temperature of 65°F, the following tests shall be accomplished:

a. No load: Unwind cable to maximum length

b. 6,500-pound line pull: Rewind cable from maximum length, stopping the cable drum every 5 seconds for 100 feet of cable travel. Then subject the winch to five stopping operations in a period of 10 seconds. The pendant control shall be utilized in this operation. Malfunction or operational irregularity shall be cause for rejection.

4.6.8 Static load test. The winch shall be subjected to a static load of 9,700 pounds through the cable at the conclusion of all preproduction tests. There shall be no permanent deformation to any part of the winch, excluding cable, during the test.

4.6.9 Final disassembly and inspection. After the static load test, the winch shall be disassembled and all parts inspected for distortion, excessive wear, and other unsatisfactory conditions.

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4.6.10 Reliability demonstration. A reliability demonstration test shall be performed on two complete winches selected at random by the Government from the first 20 winches produced to demonstrate a 50-hour MTBF in accordance with MIL-R-27542 and the performance requirements outlined in section 3 herein. Minimum test time on each winch shall be 56 hours for a total test time of 112 hours. A Chi-Squared distribution test shall demonstrate this 50-hour MTBF at 90 percent confidence.

4.6.10.1 Mission. A 15-minute mission is defined in four methods, each having different operations, but defining a typical 15-minute mission. Testing shall be distributed between methods by percentage as indicated below.

4.6.10.1.1 Method 1 - 15 percent of total test time. With a 6,500-pound line pull, the winch shall unreel and reel in 100 ±10 feet of cable two times, stopping four times for 10 seconds each, at intervals of 50 feet of cable travel.

4.6.10.1.2 Method 2 - 25 percent of total test time. With a 3,000-pound line pull, the winch shall unreel and reel in 100 ±10 feet of cable two times, stopping four times for 10 seconds each, at intervals of 50 feet of cable travel.

4.6.10.1.3 Method 3 - 40 percent of total test time. With a 1,000-pound line pull, the winch shall unreel and reel in 100 ±10 feet of cable one time, stopping four times for 10 seconds each, at intervals of 50 feet of cable travel.

4.6.10.1.4 Method 4 - 20 percent of total test time. With any line pull up to 200 pounds, and the drum in free travel mode, unreel 200 feet of cable. Check unwind limit switch. Rewind 200 feet of cable stopping every 25 feet. Check overwind limit switch.

4.6.10.2 Failure. A failure is defined as any event, other than normal servicing and adjustment that would prevent the winch from accomplishing the mission defined herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be levels A and C as specified.

5.1.1 Level A. The winch shall be preserved and packaged to method IIA in accordance with MIL-P-116. The winch cable shall be unreeled and all surfaces coated with type P-1 preservative as specified in MIL-P-116. All exposed unpainted metal surfaces of cable drums, sheaves, and snatch blocks shall be coated with type P-1 preservative.

5.1.2 Level C. The winch shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing

5.2.1 Levels A and B. Winches preserved and packaged as specified in 5.1.1 shall be packed in a box conforming to PPP-B-621, class 2, style optional; PPP-B-601, overseas type, style optional; or in a box conforming to MIL-B-26195, type 1, style and class optional.

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5.2.2 Level C. Winches preserved as specified in 5.1.2 shall be packed in containers which afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.2.3 The winch and all accessories shall be packed in one outside container.

5.3 Marking for shipment. Interior and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Winch, Aircraft Mounted, Type HCU-9/A

6. NOTES

6.1 Intended use. The Type HCU-9/A winch is intended for use in loading and unloading cargo in and out of C-141, C-130E, and C-133 aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Point of inspection.

c. Selection of applicable levels of preservation and packaging, and packing.

d. Desired electrical power cable (see 3.5.7).

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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